Homework 2
Object Tracking

TA. 單宇晟 曹博鈞 吳年茵 曾偉杰

Objective

- Implement visual multiple object tracking on videos
- Using **detection model + Hungarian algorithm**
  - **Using tracking model directly are NOT permitted**
  - Calculate the total number of people appearing in the video
- Output the tracking result video

Steps

1. Choose a **detection model** and do detection for each frame.
   - You can use pre-trained weight or train by yourself.
   - It is only necessary to detect the ‘people’ category.
2. Using **Hungarian algorithm** to match the bounding boxes.
   - You can use existing sklearn function.
   - Decide on the cost factor by yourself. (IOU, distance, ReID similarity, etc.)
   - **If same person leaves and re-enters the frame, they should be counted as a different individual.**
3. Calculate the total number of people in the video.
4. Render tracking results and save the video (requirements on the next page)

Output Video Requirements

1. **People count on the upper left corner**
2. **People bounding boxes**
   - Box of different instance should be colored with difference colors.
   - Box of the same instance should maintain the same color within frames.
   - unique box_id labeled on box.

Output Video Requirements - Bonus

- **Trajectory visualization**
  - Visualize each people's trajectory

Benchmark

- The compressed assignment file provides the following data:
  - **‘easy_9.mp4’**: A simple scene (no overlapping), used to check if tracking successfully implemented.
  - **‘hard_9.mp4’**: A complex scene (with overlapping), used to evaluate the handling of intersecting bounding boxes.
  - **’easy_9_output.mp4’**: Tracking result example of ’easy_9.mp4’, including bonus.
  - The number after the underscore in the filename represents the correct answer.
  - You can also test with other videos of your choice.
- You **must** save the video, including the requirements mentioned before.
- We will grade based on the result video.

Grading Policy

- **Online Demo**
  - We will have DEMO in **11/12**, **11/13** Via Google Meet.
    - If you can’t demo on these two days, contact TA.
  - The procedure requires you to share your screen and turn on your microphone.

    Please ensure that these functions are working properly.
  - Please go to **Google Sheet** and fill demo time you prefer.
  - The meeting link will be announced on the demo day.

Grading Policy

- **Tracking result**
  - **easy case – 60 points**
    - Show the result video of tracking **'easy_9.mp4'**.
    - 10 points will be deducted for each error.
  - **hard case – 20 points**
    - During the DEMO, you will get a video **'demo.mp4**'.
    - Run your tracking procedure and display the tracking result video.
    - You can have up to 3 errors, and 3 points will be deducted for each additional error.
  - **Bonus: Trajectory visualization – 10 points**
  - **Definition of error**
    - **People count mismatch**
      - e.g. predict: 4, answer: 7, and there're abs(7- 4) =3 errors
    - **Wrong tracking** (box id switch**,** inconsistent individual-bbox mapping)
      - e.g. 2 people swap bounding boxes after overlapping, 1 times = 1 error

Grading Policy

- **Q&A – 20 points**
  - TA will ask some questions about your implementation.
- **If you don't implement using detection model + Hungarian algorithm, will get 0 points**

Submissions & Penalty

- Your submission should contain **only your code**
  - Do not contain any model weight or videos
- Compress into **one zip file** named **HW2_{studentID}.zip**
- Format penalty - **20 points**
  - Submit in wrong name, wrong format, wrong file, etc.
- Late penalty - **10 points per day**
  - 1 day -> 10 points, 2 days -> 20 points, and so on.
- You can use any code from Github, but **DO NOT** copy from your classmate!
  - Any copying will result in a score of zero.
- **Deadline : 11/10 23:55**

Reference

- https://hackmd.io/@computerVision/S18nD20Vq
- https://zh.wikipedia.org/zh-tw/匈牙利算法
- https://alu2019.home.blog/2021/01/20/edge-ai-multiple-object-tracking-mot-duo-ge-wu-ti/
- https://blog.csdn.net/your_answer/article/details/79160045

Note

- If you have any questions, please contact us via the E3 system and include **ALL** TAs in your message.